

EES301: Global biogeochemical cycles and Climate change

Phosphorous Cycle

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Introduction

- Like N, P influences or controls many biogeochemical processes in biosphere.
- Its concentration/availability is low/poor in soil, which often limits plant growth.
- The atmosphere does not play any major role in the movements of P – mostly sedimentary cycle
- Redox reactions play a very minor role in controlling its reactivity & distribution
- Like N, P is also added extensively to land surface as fertilizers

Availability of P

- P exists in nature as
 - organic forms (nucleic acids, phospholipids, ATP, NADP)
 - inorganic particulate phosphates (e.g., minerals such as apatite $[\text{Ca}_5(\text{PO}_4)_3\text{OH}]$, Vivianite, Struvite, etc.)
 - dissolved inorganic phosphates
 - In oceans, some fraction of dissolved P is organic, e.g., phosphonates (about 25 % organic 'P' pool in nature) and phosphate esters
- Inorganic P found in water, soil and sediments as H_2PO_4^- or HPO_4^{2-} or phosphate ion (PO_4^{3-}) ion – reacts quickly with other ions (e.g., Fe, Al) and converts to unavailable forms (immobilization)
- Depends on pH of the environment - low pH favors solubilization

Dissolved inorganic forms of P

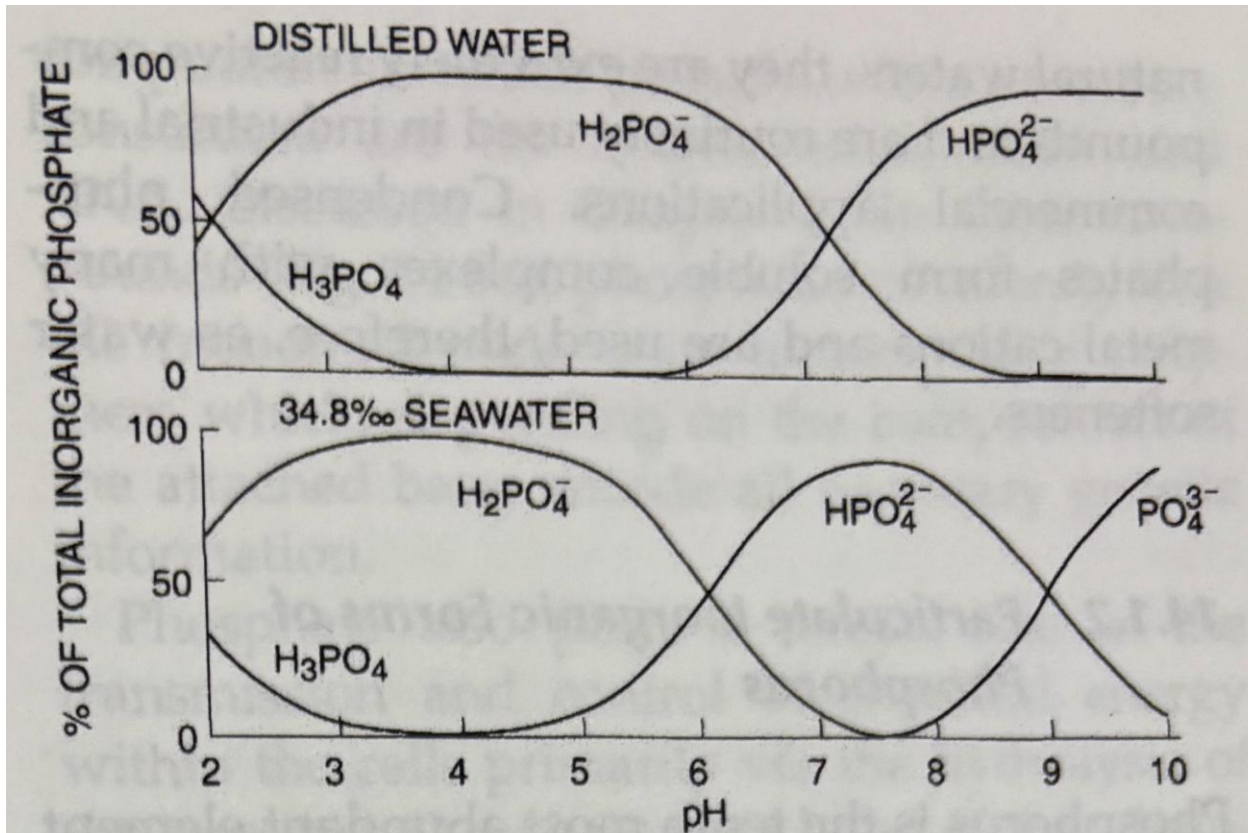
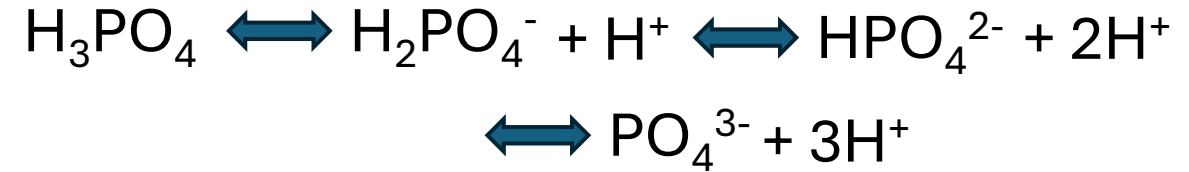


Fig. 14-1 Extent of dissolution of phosphoric acid species as a function of pH in distilled and sea waters (Atlas, 1975).



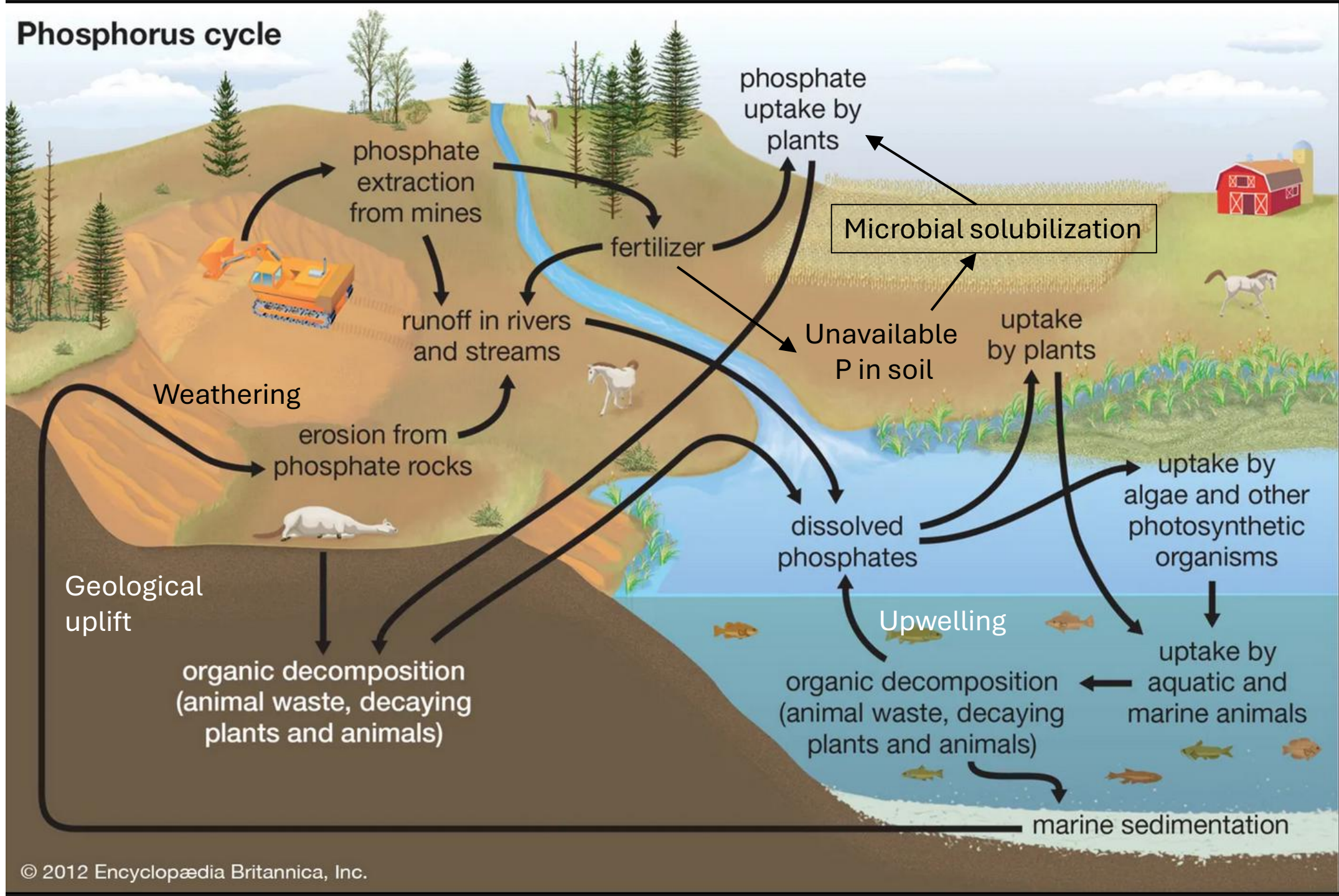
- Freshwater systems; pH 6-7, H_2PO_4^- dominates.
- Seawater; pH 8, HPO_4^{2-} dominates – PO_4^{3-} ions constitute a small proportion.

Most common oxidation states of P

- **Most phosphates (e.g., HPO_4^{2-} , PO_4^{3-}): +5**
- Phosphonates: +3
- Phosphite (H_2PO_3^-): +3
- Hypophosphite (H_2PO_2^-): +1

Nearly all dissolved and particulate P forms are combined, complexed or slightly modified forms of phosphate ion – **biogeochemical cycle thus revolves around phosphate.**

Phosphorus cycle



P moves through rocks (inorganic P), water, soil & sediments (as inorganic and organic P), & living organisms (as cellular P)

Key processes

- **Release:** Weathering & mining causes rocks to release phosphate ions (+ raining – P dust **and fertilizers**)
- **Assimilation:** of inorganic P by microbes, plants and then by animals (Organic-P)
- **Solubilization/Decomposition/mineralization** by microbes
- **Immobilization and transfer:** Precipitated P in soil ends up in waterways (through runoff and erosion) and eventually oceans. Over time, it is incorporated into sediments
- **Geological uplift/Mining**

Role of microbes in P cycling

(both sinks and sources of bioavailable P)

- **Mineralization:** Organic P compounds (e.g., phytin, nucleic acids, phospholipids) are mineralized to PO_4^{3-} by bacteria, actinomycetes and fungi
 - Phosphatase enzymes are involved in this process
- **Assimilation:** 'P' become part of cellular components
- **Immobilization:** Some microbes store P as polyphosphate granules – energy and P source

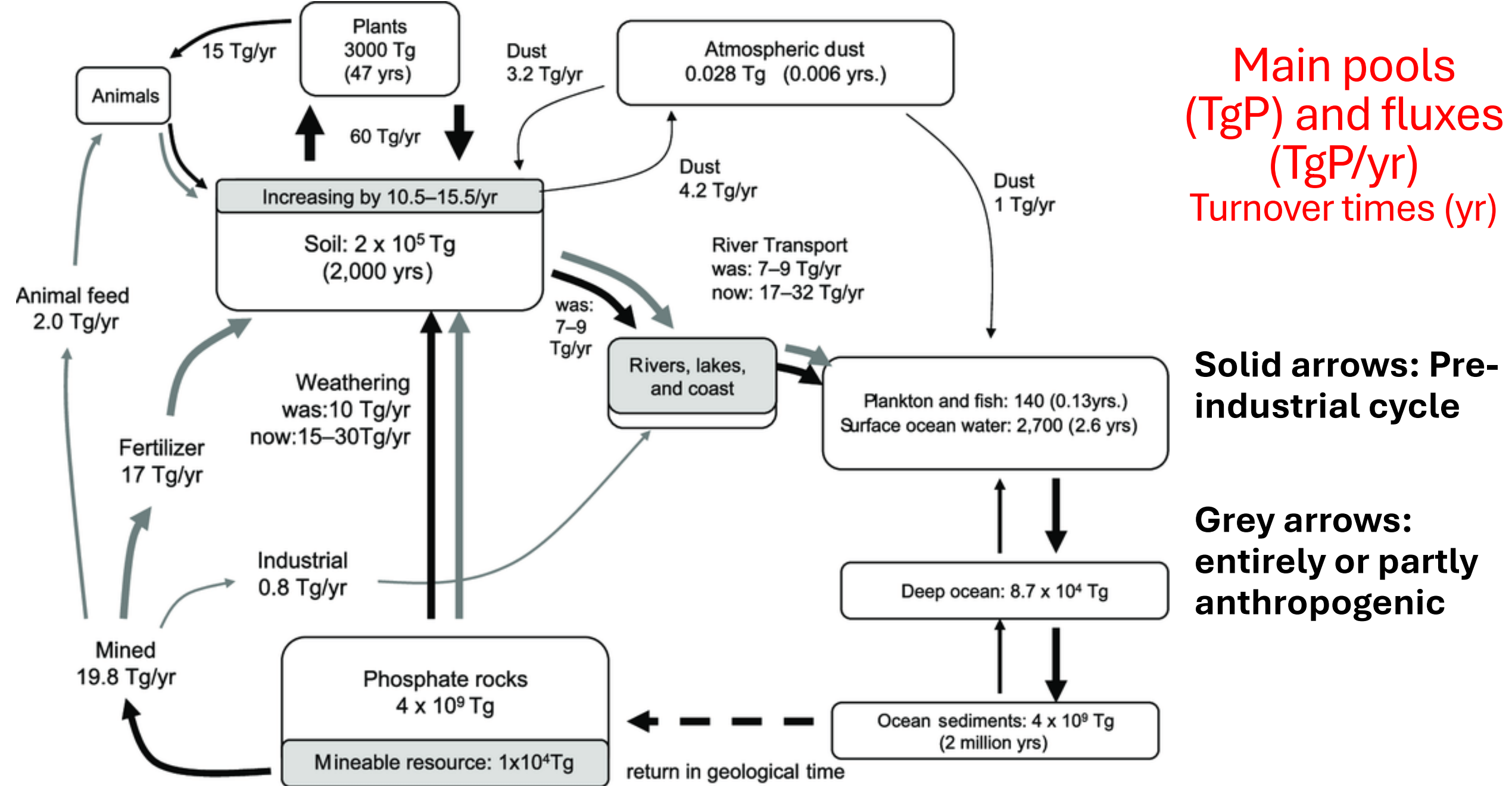
Microbial solubilization of insoluble P:

- P solubility is affected by pH and the presence of Ca^{2+} , Mg^{2+} , Fe^{3+} and Al^{3+} ions
- Solubilization through metabolic activities such as:
 - production of chelators (e.g. gluconic acid) which can leach out phosphates from complexes (Ca-, Mg- and Al-)
 - production of organic and inorganic acids (e.g., succinic & oxalic, nitric & sulfuric acids)
 - production of CO_2 : lowers pH
 - production of H_2S : may react with FePO_4 and liberate PO_4^{3-}
- Phosphate-solubilizing microbes: used as biofertilizers

Cycling of other P forms

- **Phosphonates** – organophosphate compounds (direct P-C atoms bonding); produced by some microbes - **Microbial degradation in phosphate depleted marine environments**
- Other P forms include reduced Phosphite (H_2PO_3^-) and Hypophosphite (H_2PO_2^-)
- **Phosphite oxidation coupled to sulfate reduction -**
Chemolithotrophic process yields phosphate and sulfide





**Main pools
(TgP) and fluxes
(TgP/yr)
Turnover times (yr)**

Solid arrows: Pre-industrial cycle

**Grey arrows:
entirely or partly
anthropogenic**

Human influence on the global P cycle and its impacts

- Mining – Fertilizers (P is non-renewable resource, mineable P is limited – serious concerns over its availability in the future)
- Excessive use (detergents, fertilizers, etc) and poor management – environmental pollution
- Adding more available P where little was available previously – release in the water bodies – eutrophication – dead zones